

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

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Exposé VIII. The Finiteness Theorem

1. A Biduality Spectral Sequence¹

Let us state the result we wish to establish:

Proposition 1.1. *Let A be a Noetherian ring and let I be an ideal of A . Put $X = \operatorname{Spec}(A)$ and $Y = V(I)$. Let M be a finitely generated A -module of finite projective dimension. Let $\mathcal{F} = \widetilde{M}$ be the $\mathcal{O}_X$ -module associated with M .*

1) *There is a spectral sequence*

$$H_Y(X, \mathcal{F}) \Leftarrow \operatorname{Ext}_Y^p(\operatorname{Ext}^{-q}(M, A), A).$$

2) *There is a spectral sequence*

$$\mathcal{H}_Y(X, \mathcal{F}) \Leftarrow \mathcal{E}xt_Y^p(\mathcal{E}xt^{-q}(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X).$$

¹A reader familiar with Verdier's language of derived categories will recognize the spectral sequence associated with a *biduality isomorphism*. Cf. SGA 6, I.